

(Marco) Xiaolin Zhang

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EDUCATION

Washington University in St. Louis

St. Louis, MO

Bachelor of Science in Statistics & Computer Science, GPA: 3.94 / 4.0

Aug 2024 - May 2028

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, MySQL, Go, Python, Kotlin

Web Development: React, HTML/CSS, Spring Boot, Express.js

Databases and Cloud: MySQL, PostgreSQL, Redis, GCP, AWS

PROJECTS

OnlineOrder: A Spring Boot based online food ordering system

- Developed CRUD REST APIs using Spring Controllers, encompassing functionalities such as registration, menu searching, ordering, and checkout
- Leveraged Spring Data JDBC and repositories to interface with a PostgreSQL database hosted on AWS RDS, handling data related to menus, restaurants, and more
- Implemented application authentication via Spring Security, employing session-based authentication
- Architected the project with a clear separation of concerns into controller, service, and repository layers, utilizing dependency injection for enhanced maintainability
- Constructed the frontend using React and Ant Design, providing users with the ability to seamlessly add items to their shopping cart and place orders
- Containerized and deployed it to AWS App Runner for scalability and accessibility

SocialAI: an AI based Social Network

- Designed and implemented a social network web application with React
- Used OpenAI's Dall-E 3 to assist users in creating and updating posts
- Improved the authentication using token-based registration/login/logout flow with React Router and server-side user authentication with JWT
- Launched a scalable web service in Go to handle posts and deployed to Google AppEngine
- Used Elasticsearch (deployed to GCE) to provide search functions such that users can search recent posts and list personal posts

Next AI: Full-Stack Web-Based Q&A AI Agent for PDF Document Queries

- Engineered an interactive conversational UI leveraging React and Ant Design, enabling users to upload and interact with PDF documents in real-time effortlessly
- Architected and implemented RESTful APIs via Express and Node.js, optimized for high-performance request handling. Utilized a memory vector store to cache generated embeddings for efficient retrieval
- Integrated OpenAI's GPT-5 API and Langchain technologies, resulting in a sophisticated AI Agent capable of document loading, splitting, storage, retrieval, and output functionalities

EXPERIENCES

Teaching Assistant: CSE 2407 Data Structure and Algorithms; CSE1302 Introduction to Computer Engineering; CSE 3302 Object Oriented Studio

- Led Studios, Assisted and Supported like-aged students in mastering core data structures and algorithms in Java, basic computer architecture knowledge, and C++

WashU Formula SAE - Team of Electronic and Data Acquisition

- Designed, Implemented, and Optimized the race car's electronic and wiring system
- Designed Sensor Board using trending industrialized software Altium
- Managed the whole race car's wiring harness with Rapid Harness